

Referee response for “Higgs-mode dynamical magnetoelectric effect in a van der Waals multiferroic”

Referee 1

Summary of the Work

The manuscript reports on a study of time-resolved Second Harmonic Generation (SHG) performed on the layered multiferroic compound CuBr₂. The authors observe that SHG intensity oscillates following irradiation with femtosecond pulsed laser at a wavelength of 1300 nm or 1650 nm. Notably, the oscillation frequency is shown to be dependent on the polarization configuration:

- Higher-frequency oscillations are observed exclusively in the crossed polarization channels.
- Slower-frequency oscillations appear regardless of the specific polarization settings.

The authors attribute these oscillations to the spin system. Specifically, they argue that the SHG signal monitors ferroelectric polarization induced by the Cu spin cycloid. They rule out phonon-driven oscillations on the basis that known/calculated phonon frequencies are significantly higher than those observed. Ultimately, the authors ascribe the higher-frequency and lower-frequency oscillations to the anisotropy-gapped Goldstone mode and the Higgs mode of the spin system, respectively.

Referee comment

Correlation Between SHG Signal and Ferroelectric Polarization:

I am not yet convinced that the SHG signal definitively monitors ferroelectric polarization as proposed. In the present case, where the ferroelectric polarization vector (parallel to the y-axis) lies within the incident plane, the SHG signal should theoretically appear only in the P-out channel, regardless of the incident light polarization. Consequently, the P-in/S-out and S-in/S-out channels should exhibit SHG minima at similar incident plane orientations. However, Figure 4 appears to contradict this:

- A strong SHG intensity is observed at approximately $\phi = \pi/2$ for the P-in/S-out channel.
- Conversely, the intensity peaks at approximately $\phi = 0$ for the S-in/S-out channel.

The authors should clarify this discrepancy to validate their interpretation of the SHG origin.

Our response

Referee comment

Interpretation of the Slower Oscillation as a Higgs Mode:

I question whether the observed slower oscillation can be strictly attributed to a Higgs mode (i.e., an amplitude fluctuation). If the ordered cycloid component were fluctuating, one would expect the ferroelectric polarization to both rise and fall. However, the data shows that the SHG signal repeatedly decreases and recovers, but does not increase beyond its baseline. The authors should address why the expected "rise" in polarization is absent if this is indeed an amplitude fluctuation mode.

Our response

Referee comment

Experimental Configuration:

The description of the experimental geometry is currently unclear. In the caption for Figure 2, the authors state: "The sample normal is perpendicular to the (001) lattice plane, with the $\phi = 0$ axis coincident with the chain direction in Fig. 1a". Please clarify what is meant by the " $\phi = 0$ axis" and whether it is distinct from the ϕ axis itself.

Our response

Referee comment

Symmetry and Orientation:

Since the sample normal is the c^* -axis but not the twofold axis (b-axis) of the monoclinic crystal, the sample orientation is not uniquely determined solely by the orientation of the b-axis (chain direction).

Our response

Referee comment

Incident Angle:

The incident angle of the fundamental light, θ , must be explicitly stated. Specifically, is θ larger than the angle between the y-axis and the c*-axis? This relationship significantly impacts the resulting SHG signals and their interpretation.

Our response

Referee comment

Contextualizing Magnon Excitation:

On page 1, the authors state: “Conventionally, the DME is realized by coupling an external electric field to the electric dipole active pseudo-Goldstone magnon”. It should be noted that various types of magnons can be excited by oscillating electric fields. For instance, several types of electromagnons were identified in perovskite rare-earth manganese oxides over a decade ago (e.g., N. Kida et al., J. Opt. Soc. Am. B 26, A35 (2009)).

Our response

Referee comment

Predicted Signal for Goldstone Mode:

If the anisotropy-gapped Goldstone mode of the spin cycloid is excited, the orientation of the ferroelectric polarization should oscillate. In principle, the magnitude of the polarization in its original orientation (along the y-axis) should decrease at a frequency twice as large as the oscillation, while an oscillating electric polarization should appear along the z-axis. It is unclear why such a signal is not observed in the S-in/S-out or P-in/P-out channels.

Our response

Referee comment

Supporting Information:

The measurement temperatures must be explicitly labeled in Figure S1.

Our response

Referee 2

The work presented by Fichera et al picks up two contemporary aspects in research on quantum materials. First, after the magnetoelectric correlations of multiferroics have been investigated for quite some time now, dynamic magnetoelectric correlations move into focus. Second, the magnetic and electric properties may become tightly entangled through fluctuations and criticality in the quantum world with a potential for novel quantum effects, especially in the low-THz frequency regime.

On this background, the authors of the present work report the observation of magnetoelectrically active Higgs modes in a layered multiferroic. Normally, these would decay immediately, but in the special combination of quantum criticality, mode energies, and spin-charge interaction they may become long-lived and thus observable.

While the observation and discussion of the author's claims in the manuscript are quite qualitative, raising the question if explanations alternative to Higgs-mode observation may be possible, the Supplementary Information presents supporting quantitative analysis. In total, I can imagine publication of this work in Nature Materials if my following concerns can be ruled out.

Referee comment

(1) I do not understand the statement on page 2 according to which a purely magnetic c-type electric-dipole term is not permitted by the spiral order, implying it is caused by the induced electric polarization. Since the polarization is improper, induced by the magnetic order (Eq. 2), the symmetry of the magnetic system should be equal or lower than that of the polar order. This, however, would contradict the quoted statement.

Our response

Referee comment

(2) Another statement on page 2 is that the Higgs mode should modulate all the SHG susceptibility components equally. This would be true if they are all magnetically induced. However, we clearly have some contributions not coupling to the magnetic order, as the nonzero signal in Fig. 2c above the critical temperature shows. Furthermore, the polarization rotation mentioned at the top of page 6 would mix susceptibilities and modulate them to a different degree even if there were no “non-magnetic” SHG background.

Our response

Referee comment

(3) I do not see where the van-der-Waals nature of the material studied here matters for the Higgs-mode observation. Also, these systems are known for their novel ferroelectric sliding modes that may occur in the frequency range investigated here. What is their role in the game?

Our response

Referee comment

(4) Formal mistakes: Insets and labeling in Fig. 3 are much too small. The “dashed lines” in Fig. 4 are dotted lines in my printout.

Our response

Referee 3

This work provides evidence for a Higgs mode that is stabilized by proximity to a quantum critical point. Dynamical measurements in CuBr_2 detect a low-energy mode that is surprisingly long lived. The authors interpret this mode terms of a Higgs mode, which is made long-lived because a proximity to a quantum critical point makes the Higgs mode to lie below the Goldstone mode that is made to have a nonzero energy due to magnetic anisotropy. Theoretical speculations along this line have existed before (eg Ref 23).

Still, the experiment is striking and the mechanism brings out the general conceptual importance, viz new dynamical effect is enabled by quantum criticality. The fact that this general physics is happening in a class of materials that have not in the past been highlighted in the context of quantum criticality further elevates the importance. I therefore feel that the work is potentially suitable for publication in a high profile journal like Nature Materials.

Referee comment

The interpretation rests on CuBr_2 to be born nearly quantum critical. Is there any evidence—thermodynamic or otherwise—for this in the material?

Our response

Referee comment

Second, to say that the authors have really “uncovered” a Higgs-mode dynamical magnetoelectric effect seems to me to be rather strong. It seems to me that providing evidence for the effect, or some such weaker statement, is more suitable. This will not diminish the conceptual importance of the work, but will add motivation for further studies along this general direction.

Our response

Referee comment

Third, I would like to see remarks about whether this is the first material in which evidence has been provided for the Higgs mode energy to lie below the Goldstone mode energy, or whether related effect has been observed in other materials classes – such as magnets. This again does not take away the freshness of the observed effect in the current context of multiferroic materials for which the relevance of quantum criticality has been considered to be rare.

Our response

References